Designer Genes C - Designer Genes - Monta Vista Science Olympiad (Div. C) - 11-14-2025
Student Answer Sheet (/monta-vista/SM/AnswerSheet?tid=000GFM)

Instructions (shown before students start the test)

Introduction (shown after students start the test)

Section 1: The Ideas of Mendel

1. (1.00 pts) Mendel's Law of ______ states that every individual organism contains two alleles for each trait and they seperate during meiosis

2. (1.00 pts) During Mendel's experiments, he idealized a 3:1 ratio between dominant and recessive phenotypes. But in a dihybrid cross, he found a ratio of:

- A) 3:1
- B) 1:1:1:1
- C) 3:1:1:3
- D) 9:3:3:1

3. (1.00 pts) In Labrador retrievers, coat color is determined by two genes:

Gene B = pigment color (B = black, b = brown)

Gene E = pigment deposition (E = allows color, e = no pigment → yellow)

What is the expected phenotypic ratio when two heterozygotes (BbEe × BbEe) are crossed?

- A) 9 black: 3 brown: 4 yellow
- B) 12 black : 3 brown : 1 yellow
- C) 9 brown : 3 yellow : 4 black
- D) 15 black : 1 yellow

4. (1.00 pts) Which statement best describes the implication of epistasis for genetic studies?

- A) It supports Mendel's law of segregation but not independent assortment.
- B) It shows that some traits are controlled by environmental factors only.
- C) It demonstrates that genes can interact in pathways, modifying Mendelian ratios.
- D) . It implies that only dominant alleles are expressed.

Section 2: The two M cycles

5. (1.00 pts) The main purpose of meiosis is to produce genetically identical cells

○ True ○ False

6. (1.00 pts) Crossing over occurs during prophase I of meiosis

○ True ○ False

7. (1.00 pts) Spindle fibers are unique to meiosis

○ True ○ False

8. (1.00 pts) If nondisjunction occurs during Meiosis I, what happens?

○ A) Two gametes are normal, two abnormal
○ B) All gametes are abnormal
,
○ C) no effect
○ D) Results in polyploidy

9. (1.00 pts) Which of the following events occurs in both mitosis and meiosis II?

○ A) Separation of homologous chromosomes
○ B) Separation of sister chromatids
○ C) Crossing over
○ D) Synapsis

10. (3.00 pts)
A 35-year-old woman undergoes prenatal genetic screening. The doctor performs a karyotype on fetal cells and finds 47 chromosomes, including three copies of chromosome 21.

Answer the following based off that information:

Explain what error during meiosis could have caused this result.
|(b) Identify the name of this condition and describe how it affects chromosome number in the gamete and zygote.

(c) Describe how this error differs from a mutation that occurs during mitosis.

Section 3: The Depths of DNA

11. (1.00 pts) The sugar in DNA lacks which functional group found in RNA?

- A) 3′-OH
- B) 2′-OH
- C) 5′-OH
- D) Carbonyl Group

12. (1.00 pts) The 5′ end of a DNA strand is characterized by a:

- A) Free hydroxyl group on carbon 3′
- B) Free phosphate on carbon 5'
- C) Nitrogenous base
- D) Deoxyribose

13. (2.00 pts) Explain the formation of Phosphodiester bonds

14. (1.00 pts) Which of the following levels of DNA organization is the loosest and most transcriptionally active?

- A) Chromosome
- B) Heterochromatin
- C) Euchromatin
- D) Nucleosome

15. (1.00 pts) DNA in nucleosomes is wrapped around:

- A) Histone octamers
- B) Actin filaments
- C) RNA particles
- D) Ribosomes

16. (1.00 pts) A point mutation changing an A→G in the third position of a codon but coding for the same amino acid is a called ____ mutation. (Hint the answer is 1 word)

17. (1.00 pts) A mutation that converts a codon for an amino acid into a stop codon is a ___ mutation

18. (2.00 pts) Why is the mutation in the previous question so harmful? What's an example of its effect?

19. (1.00 pts) Which of the following can cause DNA damage?

- A) X-rays
- B) Gamma rays
- C) Chemical exposures
- D) All of the above

20. (2.00 pts) Pyrimidine Dimers is a type of DNA damage caused by:

- A) Bacterial toxins
- B) UV radiation
- C) Errors in DNA replication during S phase
- D) Ioniziation

21. (1.00 pts) Which DNA repair mechanism specifically removes and replaces incorrectly paired bases that escape proofreading during replication?

- A) Nucleotide excision repair (NER)
- B) Mismatch repair (MMR)
- C) Base excision repair (BER)
- D) Homologous recombination (HR)

22. (1.00 pts) Nucleotide excision repair (NER) primarily fixes which type of DNA damage?

- A) Double-stranded breaks from X-rays
- B) Thymine dimers
- C) Depurination from loss of bases
- D) Mispairs from replication

23. (1.00 pts) DNA polymerase can synthesize new DNA strands in both the 5′→3′ and 3′→5′ directions.

True False

24. (1.00 pts) Hydrogen bonds between complementary bases stabilize the DNA double helix.

True False

Section 4: How the prokaryotes do it

25. (1.00 pts) Which enzyme catalyzes transcription in prokaryotes?

- A) DNA polymerase III
- B) RNA Polymerase
- C) Ligase
- D) Reverse Transcriptase

26. (1.00 pts) The σ (sigma) factor in bacterial RNA polymerase is responsible for:

- A) Elongation
- B) Termination
- C) Ribosome binding
- D) Promoter recognition and initiation

27. (1.00 pts) The lac operon is activated when lactose is present and glucose is low

True False

28. (1.00 pts) In the trp operon, tryptophan acts as a inducer

True False

29. (1.00 pts) In prokaryotes, transcription and translation can occur simultaneously.

True False

30. (1.00 pts) Riboswitches are regulatory RNA segments that can bind metabolites and alter gene expression.

True False

31. (1.00 pts) The Shine-Dalgarno sequence is found on the 3′ end of mRNA.

True False

32. (2.00 pts) The codon that signals the start of translation is usually __________, which codes for the amino acid __________ (FULL AMINO ACID NAME)

33. (1.00 pts) The sequence on bacterial mRNA that helps the ribosome identify the correct start codon is the __________ sequence.

34. (1.00 pts) The DNA sequence where a repressor binds to block transcription in an operon is called the __________

35. (3.00 pts) The lac operon is a pretty important operon in prokaryotes. Answer the following about it

(a) Describe the roles of the lac repressor (LacI) and the CAP-cAMP complex in regulating the lac operon.

(b) Explain what happens to lac operon transcription when lactose is present but glucose is high.

(c) Predict the effect of a mutation that deletes the operator region and explain why.

Section 5: The Techniques

36. (1.00 pts) During the denaturation step of PCR, which bonds are broken?

- A) Covalent bonds between sugars and phosphates
- B) Hydrogen bonds between complementary bases
- C) Disulfide bonds in proteins
- D) Amino acid peptide bonds

37. (1.00 pts) What is the typical temperature range for the denaturation step in PCR?

- A) 40-50°C
- B) 94–98 °C
- C) 25–30 °C
- D) 55–65 °C

38. (1.00 pts) Which enzyme is used in PCR due to its heat stability?

- A) DNA polymerase III
- B) RNA polymerase
- C) Taq polymerase
- D) Ligase

39. (1.00 pts) Where does that previously mentioned enzyme come from?

40. (3.00 pts) There's some cool techniques used in biotechnology! Answer the following:
(a) Describe the role of temperature in each step of the PCR cycle (denaturation, annealing, and extension).

(b) Explain how Sanger sequencing uses modified nucleotides to determine DNA sequence.

(c) Identify one biological question/idea that can be answered using PCR, and one that can be answered using Sanger sequencing.

41. (1.00 pts) How many copies of DNA are theoretically produced after 5 PCR cycles (starting from 1 molecule)? (Just put in a number as your answer, for instance '67')

42. (1.00 pts) PCR is commonly used to:

- A) Detect pathogens
- B) Amplify DNA for cloning
- C) Analyze genetic mutations
- D) All of the above

Good job! Good luck on any other events you may have!